HIRUSHA NILUMINDA

Electronics & Telecommunication Engineering Technology Diploma Student – ITUM

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PROFILE

I am an Electronics and Telecommunication Engineering student at the Institute of Technology, University of Moratuwa, with hands-on experience in embedded systems, microcontroller programming, sensor interfacing, and electronics prototyping. Experienced in developing Arduino- and ESP32-based projects involving automation, IoT monitoring, and robotics. Interested in robotics R&D, embedded systems, firmware development, PCB design, and industrial automation, with a focus on practical engineering applications.

EDUCATION

Institute of Technology, University of Moratuwa (ITUM)

National Diploma in Technology (NDT) - Electronics & Telecommunication Engineering | Present

WP/ KE/ Henegama Central College

G.C.E. Advanced Level - Physical Science | 2022

3 C passes

TECHNICAL SKILLS

- **Embedded & Programming:** Arduino, ESP32, ESP8266, STM32, Raspberry Pi Pico W
- **Robotics & Control:** ROS 2 (Currently Learning), Webots Simulation(Currently Learning)
- **Firmware & Software:** C/C++, Python (Basic), Dart (Basic), Arduino firmware development
- **Electronics:** Soldering, Sensor interfacing, Motor control
- **IoT & Connectivity:** Blynk IoT, Google Firebase
- **EDA & CAD Tools:** KiCad, Fritzing, SolidWorks, Visual Studio Code, Arduino IDE

PROJECTS

Multi-Sensor IoT Weather Station

Personal Project | ESP32-Based Monitoring and Data Logging

- Developed a breadboard prototype using an ESP32, environmental sensors, an SD-card data logging system, and a 16×2 LCD.
- Integrated a Blynk dashboard for real-time remote monitoring of weather data.
- Completing the hardware implementation on a dot board

4-DOF Robotic Arm – ESP32-Based Dual Control System

Personal Project | ESP32-Based 4-DOF Robotic Arm

- Assembled a 4-DOF robotic arm using a mechanical kit.
- Implemented wireless robotic arm control through an ESP32-based web dashboard.
- Implemented an alternative manual control system using a custom remote with two joysticks.

4 Wheel Obstacle Avoiding Robot Car

Personal Project | Arduino-Based Robotic car

- Built a four-wheel robotic vehicle with basic obstacle detection.
- Implemented manual wireless control using a custom remote controller.

Autonomous Soil Nutrition Mapping Rover

Team Project | University Final Project

- Developing an agricultural rover with custom firmware for automated soil nutrient mapping.
- Working on automated sensor insertion mechanisms for soil measurements.
- Designing communication between two microcontrollers to coordinate sensing and rover operations.

Autonomous Agricultural Robot

Team Project | Participation, INCO Competition – 2026

- Developed an ESP32-based autonomous agricultural robot for precision seed dispensing, controlled fertilizer application, and automated watering.
- Integrated ultrasonic and Hall-effect sensors for obstacle detection, wheel movement monitoring, and dead-reckoning navigation.
- Designed and built a custom remote controller to enable manual operation and control of the robot.

Eco Smart Greenhouse - Vertex 2025

Team Project | 2nd Place, Vertex 2025 Inter University Innovation Competition

- Developed embedded control software and completed system wiring using an ESP32 expansion board.
- Implemented automated environmental control using DHT11, soil moisture, and light sensors to manage pumps, fans, misting, heating, and grow lights through relay-controlled outputs.
- Integrated a Firebase dashboard for remote monitoring and an ESP32-CAM with Edge Impulse for plant disease detection.

CERTIFICATIONS & ACHIEVEMENTS

- **PLC Certificate Course** – Sri Lanka Institute of Robotics
- **From Schematic to PCB: Layout Fundamentals** – Coursera
- **Python for Beginners** – University of Moratuwa (DP Education)
- **Web Design for Beginners** – University of Moratuwa (DP Education)
- **2nd Place - Vertex 2025 Inter University Innovation Competition**, Eco Smart Greenhouse (Team InnoX)

REFERENCES

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